

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI -590 014



**Flipped class room activity report on
Design of Inverting Amplifier circuit using TinkerCAD
simulation tool**

**DEPARTMENT
OF
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted by

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S.J.M VIDYAPEETHA ®

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE,
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NAAC Accredited with 'B⁺⁺' Grade

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1. Introduction

An **inverting amplifier** is one of the most commonly used operational amplifier (op-amp) configurations in analog electronics. In this configuration, the input signal is applied to the inverting terminal of the op-amp, while the non-inverting terminal is grounded. The output signal is amplified and **phase-inverted by 180°** with respect to the input signal.

In this project, an **inverting amplifier circuit** is designed and simulated using the **TinkerCAD virtual circuit design tool**. TinkerCAD provides a safe and interactive environment to understand the working of electronic circuits without the need for physical components. The project helps in visualizing voltage amplification, phase inversion, and the effect of resistor values on gain.

2. Objectives

The main objectives of this project are:

- To understand the working principle of an **inverting amplifier**
- To design an inverting amplifier using an operational amplifier
- To simulate the circuit using **TinkerCAD**
- To observe voltage amplification and phase inversion
- To analyze the input and output waveforms
- To gain practical knowledge of op-amp applications
- To study the effect of feedback resistor values on gain

3. Software Requirements

The software requirements for this project are:

- TinkerCAD Circuits (Web-based tool)
- Web Browser (Google Chrome / Microsoft Edge / Firefox)
- Stable Internet Connection
- Optional: Google Account for login

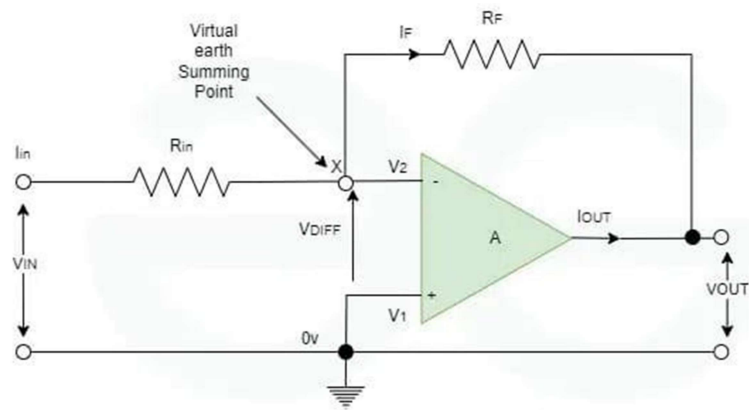
Note: *No installation is required as TinkerCAD is an online platform.*

4. Steps to Access TinkerCAD

Follow the steps below to access and use TinkerCAD:

1. Open a web browser.
2. Go to <https://www.tinkercad.com>
3. Click on Sign In (use Google account or create a new account).
4. After login, select Circuits from the dashboard.
5. Click on Create New Circuit.
6. From the components panel, select:
7. Connect the components as per the inverting amplifier circuit diagram.
8. Click on **Start Simulation** to run the circuit.

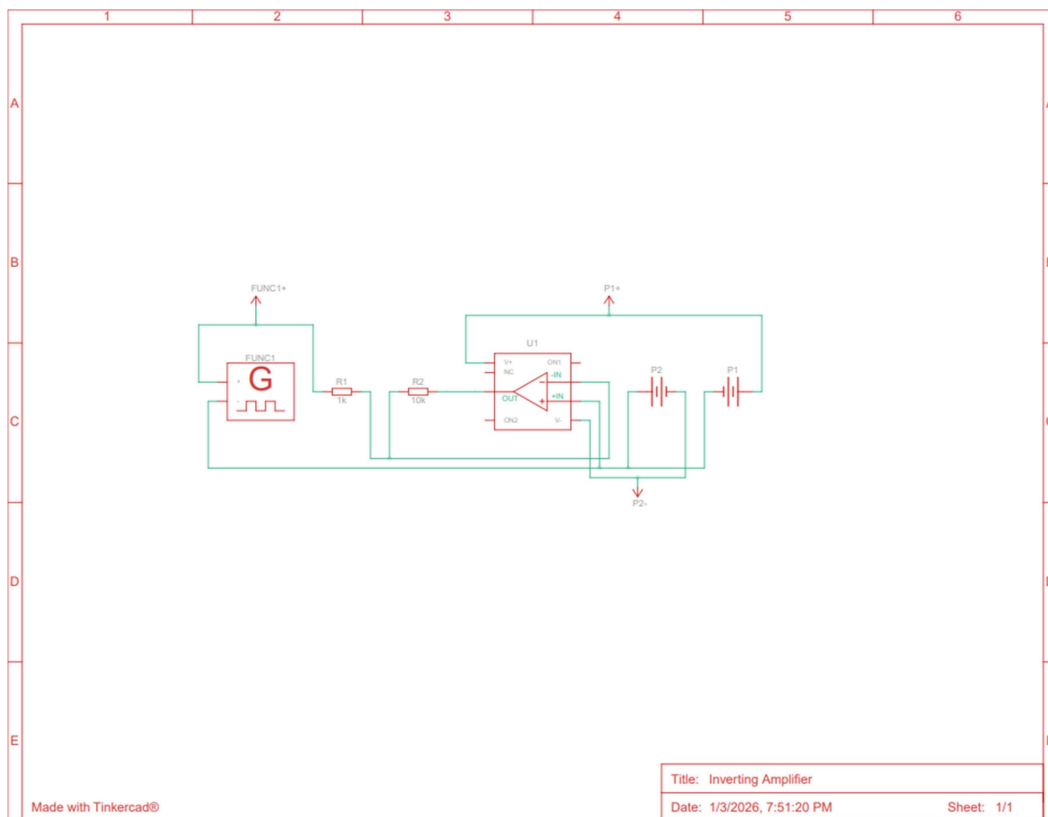
Circuit Diagram



Inverting Operational Amplifier

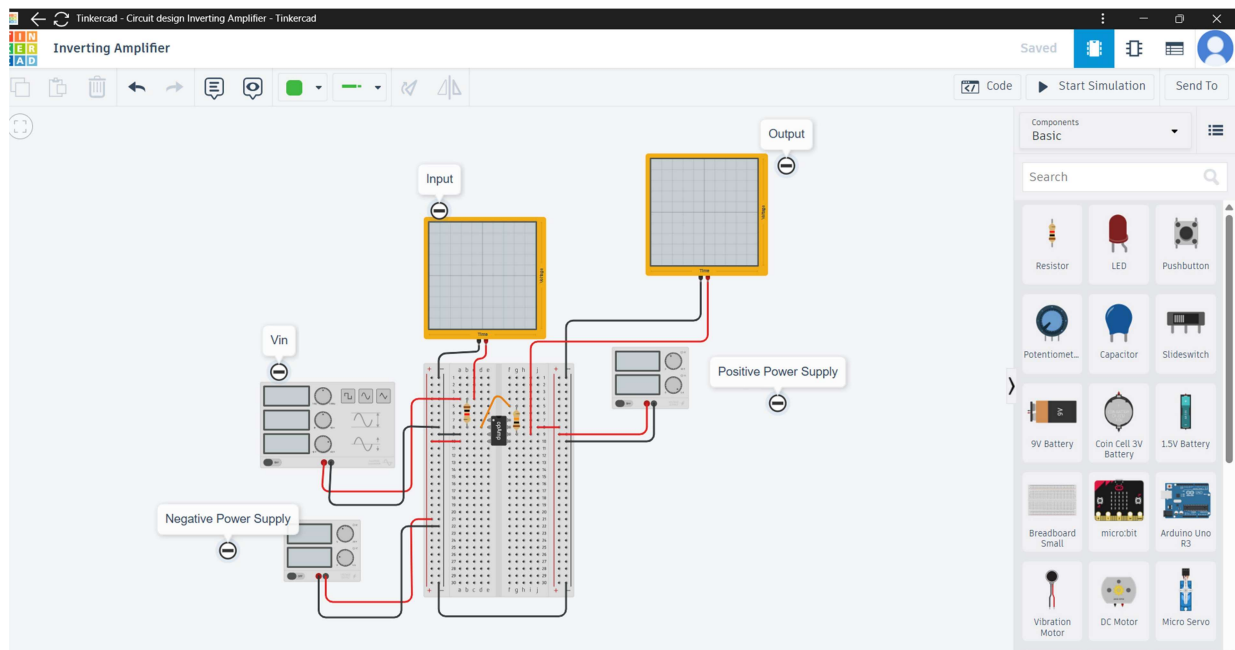


Schematic Diagram



Activity- Flipped Classroom

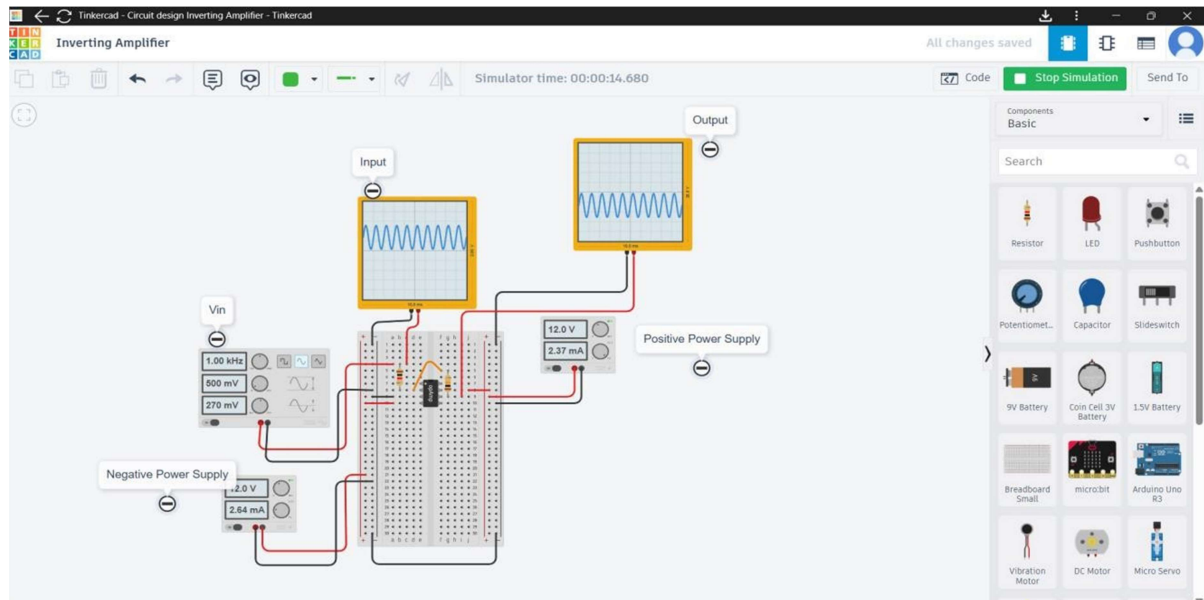
Circuit View



Components Required

Component List			Download CSV
Name	Quantity	Component	
U1	1	741 Operational Amplifier	
R1	1	1 kΩ Resistor	
R2	1	10 kΩ Resistor	
FUNC1	1	1000 Hz, 0.5 V, 0.25 V, Sine Function Generator	
U2 U3	2	1 ms Oscilloscope	
P1 P2	2	12 , 25 Power Supply	

Circuit During Simulation



5. Observation

During the simulation, the following observations were made:

- A **50 mV , 1 kHz , sine wave** is applied as the input signal.
- The circuit operates with a **500 mV DC supply**.
- When an AC input signal was applied to the inverting terminal, an amplified output signal was obtained.
- The output waveform was **180° out of phase** with the input waveform.
- The output amplitude depended on the ratio of feedback resistor to input resistor.
- A stable output was observed due to proper feedback.
- The oscilloscope clearly displayed both input and output waveforms.

6. Result and Analysis

The inverting amplifier circuit was successfully designed and simulated using TinkerCAD. The output voltage was found to be amplified according to the gain formula:

$$\text{Voltage Gain} = - (R_f/R_{in})$$

- The **negative sign** confirms the phase inversion
- The gain increased when the feedback resistor value was increased.
- The op-amp operated in its linear region, indicating correct biasing.
- The results obtained in simulation matched the theoretical calculations.

Thus, the inverting amplifier performed as expected and validated the theoretical concepts.

7. Conclusion

- The project successfully demonstrated the **practical implementation of an inverting amplifier** and verified its theoretical behavior through simulation.
- The **180° phase inversion** observed in the output waveform confirms the fundamental working principle of an inverting op-amp configuration.
- The use of **negative feedback** helped maintain circuit stability and ensured linear amplification, reducing distortion in the output signal.
- TinkerCAD provided a **risk-free and cost-effective environment** to test and analyze electronic circuits without the need for physical components.
- The project improved understanding of how **resistor values control voltage gain**, highlighting the importance of component selection in amplifier design.
- The simulation results closely matched **theoretical calculations**, validating the accuracy of the design and the effectiveness of the virtual tool.
- This project enhanced **hands-on learning experience**, bridging the gap between theoretical knowledge and practical application.
- The inverting amplifier designed in this project forms the **foundation for advanced op-amp applications** such as filters, signal conditioners, and summing amplifiers.
- Overall, the project strengthened confidence in designing, simulating, and analyzing analog circuits using modern simulation tools.